MATERIAL	SPEC NUMBER	REVISION	EFFECTIVE DATE
AISI 4140, 125K, Low Alloy Steel Non-NACE	**CMS-107**	**H**	**02/20/2023**

Material Requirement

Scope

AISI 4140, 125K Minimum Yield Strength, Low Alloy Steel, Non-NACE

Product Forms

Bar and Mill Shapes, Ring Forgings, Open and Closed Die Forgings

General Requirements and References

- CMS-002, Material Requirements for Forgings, Bars and Mill Shapes (doesn't apply for heat treatment and mechanical properties)
- ASTM E45, Standard Test Methods for determining the Inclusion Contents of Steel
- ASTM E1268, Standard Practice for Assessing the Degree of Banding or Orientation of Microstructures Note: Severe banding as shown in Figure A1.20 of ASTM E1268 is not permitted.
- API Specification 6A, Specification for Wellhead and Christmas Tree Equipment, Annex M
- SAE AMS-H-6875A, Heat Treatment of Steel Raw Materials

NOTE: The latest edition the referenced document applies unless a revision date is listed.

Chemical Composition

Element[1,2]	Composition Limits	Element	Composition Limits
Carbon	0.38 - 0.43	Silicon	0.15 - 0.32
Manganese	0.75 - 1.00	Chromium	0.80 - 1.10
Phosphorous	0.025 Maximum	Molybdenum	0.15 - 0.25
Sulfur	0.025 Maximum	Nickel	0.50 Maximum
Vanadium	0.05 Maximum	Iron	Balance
Copper	0.25 Maximum		

Note 1: Elements not listed shall not be intentionally added without the permission of Cactus Wellhead Engineering.

Note 2: Non-specified residual elements, if analyzed shall not exceed 0.05% each or 0.25% combined.

Mechanical Properties

Properties	Range
Tensile Strength, psi[3]	145,000 (1000 MPa) Minimum
Yield Strength, psi, 0.2% Offset[3]	125,000 (862 MPa) Minimum
Elongation in 2" Gage Length	12% Minimum
Reduction in Area	30% Minimum
Hardness	293-327 HBW (30.9 – 35 HRC)
Charpy V-Notch Impact properties - Test Temperature[4]: 0°F (-18°C)	
Longitudinal Toughness[5,6,7]	15/10 (20/14) Ft-lbs (J) Min. Avg/Min.
Transverse Toughness[5,6,7]	15/10 (20/14) Ft-lbs (J) Min. Avg/Min.

Note 3: The yield strength to ultimate tensile strength ratio shall not exceed 0.90.

The QTC must be a full section prolongation or a sacrificial part.

<table>
<tr><td>Mechanical Properties - Continued</td><td>The longitudinal axes of tensile and impact specimens shall be taken at a mid-radius location or deeper for a solid QTC or within 1/8” of the mid-wall for a hollow QTC.
Note 4: Test temperature shall be at or below the indicated temperature.
Note 5: Impact specimens may be taken either in the longitudinal or transverse orientation.
Note 6: Impact acceptance criteria is based upon 10 mm x 10 mm specimens.
Note 7: Toughness criteria is given in terms of the required minimum average of a seat of three specimens (first number) followed by the minimum value for a single specimen (second number). Only one specimen of a set may be below the minimum average, but in no case lower than the single specimen minimum.</td></tr>
<tr><td>Melt Practice</td><td>Electric furnace melted, vacuum degassed made to a fully killed, fine grain melting practice.
Low alloy steel suppliers must guarantee that the material being supplied is capable of meeting the following microcleanliness requirements when tested in accordance with ASTM E45, Method A:
TABLE 1 - MICROCLEANLINESS<table><tr><td>Type A</td><td>Thin 2.0 / Heavy 1.5</td></tr><tr><td>Type B</td><td>Thin 2.0 / Heavy 1.5</td></tr><tr><td>Type C</td><td>Thin 2.0 / Heavy 1.5</td></tr><tr><td>Type D</td><td>Thin 2.0 / Heavy 1.5</td></tr></table></td></tr>
<tr><td>Condition</td><td>Material shall have an ASTM grain size of #5 or finer and a minimum hot work ratio of 4:1.</td></tr>
<tr><td>Preheat Treat Machining</td><td>If tubulars are made from trepanning barstock, the tubulars shall be heat treated after trepanning. Material with an ER over 8” shall be preheat treat machined to the drawing or EPS specified dimensions prior to heat treatment.</td></tr>
<tr><td>Heat Treatment Furnace Calibration</td><td>Heat treat furnaces used for QTC’s and production parts shall meet Equipment Qualifications of CMS-002.</td></tr>
<tr><td rowspan="2">Heat Treatment General</td><td>Material shall be provided in the normalized (optional), austenitize, quenched and tempered condition.
Heat treating shall be done in accordance with the methods described CMS-002, Material Requirements for Forgings, Bars and Mill Shapes. Batch heat treating shall comply with the following. Cycle times apply only to temperature monitoring.
• Method #1 - Furnace Atmosphere Thermocouple; or
• Method #2 - Contact Surface Thermocouple.</td></tr>
<tr><td>Method #1 - Furnace Atmosphere Thermocouple
This method utilizes a thermocouple (T/C) that measures the temperature of the furnace atmosphere, but not the load itself. Holding time at temperature shall begin after the furnace atmosphere T/C reaches the set point temperature. Set point temperature must be within the allowable temperature range prescribed in the applicable material specification. Unless otherwise stated in the specific material specification, hold for the following times:
Normalizing, Austenitizing, or Annealing: After set point temperature is reached, hold for a minimum of ½ hour/inch of heaviest cross section.</td></tr>
</table>

Heat Treatment	Tempering or Aging: After set point temperature is reached, hold for a minimum of 3/4 hour/inch of heaviest cross section. If a double temper is specified, the second temper shall be held for a minimum of ½ hour/inch of heaviest cross section.
General - Continued	**Method #2 - Contact Surface Thermocouple.** This method uses a T/C physically in contact with the surface of a part in the load being heat treated. Contact shall be made on the heaviest section of the part. Holding time commences after the set point temperature has been reached and shall continue for the minimum cycle time prescribed in the specific material specification. Set point temperature must be within the allowable temperature range prescribed in the applicable material specification.
	Method for continuous furnace – The QTC shall consist of a sacrificial length cut from the end of the bar and shall be from the same heat and heat treat lot as the material it qualifies.

Heat Treatment Normalize	**Atmos/Media**	**Temperature**	**Time at Temp**
	Air	1600°F - 1700°F (871°C – 927°C)	½ Hr/Inch of Max Thru Thickness. 1 Hour Min
	Still Air Cool to Below 400°F (204°C) before processing to the next process		

Heat Treatment Austenitize	**Atmos/Media**	**Temperature**	**Time at Temp**
	Air	1550°F - 1650°F (843°C - 899°C)	½ Hr/Inch of Max Thru Thickness. 1 Hour Min
	Temperature shall be at least 25F less than normalizing temperature		

Heat Treatment Quench	**Atmos/Media**	***Time/Time at Temperature***
	Water	**1**00°F (38°C) Maximum Before Quenching 120°F (49°C) Maximum at any time during the quench cycle
	Polymer/Oil	Per the quenchant manufacturer's standards Start temperature for polymer or oil quenchant shall be per the quenchant manufacturer.
	Document the start and finish temps for quench media	

Heat Treatment Temper	**Atmos/Media**	**Temperature**	**Time at Temp**
	Air	900°F – 1300°F (482°C - 704°C)	3/4 Hr/Inch of Max Thru Thickness 1 Hour Min
	Air cool or faster to room temperature		

Heat Treatment Continuous Furnace

For bars with diameters of 8" (203 mm) or less and tubing with a wall thickness of 8" (203mm) or less, continuous furnace heat treatment is an acceptable alternative to conventional batch-type heat treatment.

	Parameters	**Time at Temp**
Austenitizing Furnace	Bar temp exiting final zone	1550°F - 1725°F (843°C - 941°C)
Austenitizing Furnace Water spray quench	Min time	5 minutes (Notes)
Tempering Furnace	Bar temp exiting final zone	900°F - 1300°F

	Tempering Furnace	Min time	(482°C - 704°C) 5 minutes (Notes)

Air cool or faster to room temperature

Notes

- The bar travels through several different temperature zones which allows the bar to heat to an austenitic state.
- Prior to entering the tempering furnace, the bar is spray quenched. Maximum water temperature at start of quench is 100°F (38°C).
- The tempering furnace zone temps are then selected to produce the required tempering effect.
- The length of the furnace and travel speed determined the time spent in the austenitizing and tempering furnace and the bar diameter affects the travel speed.
- To attain the required mechanical properties and microstructure, the time in each furnace should produce the same mechanical properties as a conventional quench-and-temper heat treatment.

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Date Prepared	*February 20, 2023*	***Date Approved:***	*February 20, 2023*

Revision History Table		
Rev H	02/20/2023	Updated heat treatment section temper time note to match required time.
Rev G	02/23/2021	Raised the HBN upper limit to match that of the HRC upper limit. Changed furnace calibration reference to CMS-002.
Rev F	02/20/2020	Raised the batch austenitize temperature limit. Added tubing wall thickness limit to Continuous heat treatment section. Raised the austenitizing temperature in the continuous heat treatment section.
Rev E	09/01/2019	Modified heat treat quench section to limit max temp throughout cycle. Revised for compliance with API 6A 21st Edition.
Rev D	09/19/2018	Completed rewrite now AISI 4140 125K Low Alloy Steel with Impact Testing and Microcleanliness requirements.
Rev C	03/20/2017	Clarified heat treat condition and heat treatment requirements
Rev B	02/14/2012	Editorial changes for improvement. Polymer quench media change to align with industry heat treat standards.
Rev A	09/01/2011	Initial Release